hr-analytics

March 18, 2024

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
import warnings
warnings.filterwarnings('ignore')
```

[2]:
```
data = pd.read_csv('C:\\Users\\kishu\\Downloads\\Employee-Attrition.csv')
```

[3]:
```
data.head()
```

[3]:
```
   Age Attrition     BusinessTravel  DailyRate              Department  \
0   41       Yes      Travel_Rarely       1102                   Sales
1   49        No  Travel_Frequently        279  Research & Development
2   37       Yes      Travel_Rarely       1373  Research & Development
3   33        No  Travel_Frequently       1392  Research & Development
4   27        No      Travel_Rarely        591  Research & Development

   DistanceFromHome  Education EducationField  EmployeeCount  EmployeeNumber  \
0                 1          2  Life Sciences              1               1
1                 8          1  Life Sciences              1               2
2                 2          2          Other              1               4
3                 3          4  Life Sciences              1               5
4                 2          1        Medical              1               7

   …  RelationshipSatisfaction StandardHours  StockOptionLevel  \
0  …                         1            80                 0
1  …                         4            80                 1
2  …                         2            80                 0
3  …                         3            80                 0
4  …                         4            80                 1

   TotalWorkingYears  TrainingTimesLastYear WorkLifeBalance  YearsAtCompany  \
0                  8                      0               1               6
1                 10                      3               3              10
2                  7                      3               3               0
```

```
3                  8                       3               3               8
4                  6                       3               3               2

  YearsInCurrentRole  YearsSinceLastPromotion  YearsWithCurrManager
0                  4                        0                     5
1                  7                        1                     7
2                  0                        0                     0
3                  7                        3                     0
4                  2                        2                     2

[5 rows x 35 columns]
```

[4]:
```
pd.set_option('display.max_columns',None)
```

[5]:
```
data.head()
```

[5]:
```
   Age Attrition     BusinessTravel  DailyRate              Department  \
0   41       Yes      Travel_Rarely       1102                   Sales
1   49        No  Travel_Frequently        279  Research & Development
2   37       Yes      Travel_Rarely       1373  Research & Development
3   33        No  Travel_Frequently       1392  Research & Development
4   27        No      Travel_Rarely        591  Research & Development

   DistanceFromHome  Education EducationField  EmployeeCount  EmployeeNumber  \
0                 1          2  Life Sciences              1               1
1                 8          1  Life Sciences              1               2
2                 2          2          Other              1               4
3                 3          4  Life Sciences              1               5
4                 2          1        Medical              1               7

   EnvironmentSatisfaction  Gender  HourlyRate  JobInvolvement  JobLevel  \
0                        2  Female          94               3         2
1                        3    Male          61               2         2
2                        4    Male          92               2         1
3                        4  Female          56               3         1
4                        1    Male          40               3         1

                JobRole  JobSatisfaction MaritalStatus  MonthlyIncome  \
0        Sales Executive                4        Single           5993
1     Research Scientist                2       Married           5130
2  Laboratory Technician                3        Single           2090
3     Research Scientist                3       Married           2909
4  Laboratory Technician                2       Married           3468

   MonthlyRate  NumCompaniesWorked Over18 OverTime  PercentSalaryHike  \
0        19479                   8      Y      Yes                 11
1        24907                   1      Y       No                 23
```

```
2         2396                    6      Y      Yes                 15
3        23159                    1      Y      Yes                 11
4        16632                    9      Y       No                 12

   PerformanceRating  RelationshipSatisfaction  StandardHours  \
0                  3                         1             80
1                  4                         4             80
2                  3                         2             80
3                  3                         3             80
4                  3                         4             80

   StockOptionLevel  TotalWorkingYears  TrainingTimesLastYear  \
0                 0                  8                      0
1                 1                 10                      3
2                 0                  7                      3
3                 0                  8                      3
4                 1                  6                      3

   WorkLifeBalance  YearsAtCompany  YearsInCurrentRole  \
0                1               6                   4
1                3              10                   7
2                3               0                   0
3                3               8                   7
4                3               2                   2

   YearsSinceLastPromotion  YearsWithCurrManager
0                        0                     5
1                        1                     7
2                        0                     0
3                        3                     0
4                        2                     2
```

[6]:
```
data.shape
```

[6]: (1470, 35)

[7]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Age                       1470 non-null   int64
 1   Attrition                 1470 non-null   object
 2   BusinessTravel            1470 non-null   object
 3   DailyRate                 1470 non-null   int64
```

```
 4   Department                1470 non-null   object
 5   DistanceFromHome          1470 non-null   int64
 6   Education                 1470 non-null   int64
 7   EducationField            1470 non-null   object
 8   EmployeeCount             1470 non-null   int64
 9   EmployeeNumber            1470 non-null   int64
 10  EnvironmentSatisfaction   1470 non-null   int64
 11  Gender                    1470 non-null   object
 12  HourlyRate                1470 non-null   int64
 13  JobInvolvement            1470 non-null   int64
 14  JobLevel                  1470 non-null   int64
 15  JobRole                   1470 non-null   object
 16  JobSatisfaction           1470 non-null   int64
 17  MaritalStatus             1470 non-null   object
 18  MonthlyIncome             1470 non-null   int64
 19  MonthlyRate               1470 non-null   int64
 20  NumCompaniesWorked        1470 non-null   int64
 21  Over18                    1470 non-null   object
 22  OverTime                  1470 non-null   object
 23  PercentSalaryHike         1470 non-null   int64
 24  PerformanceRating         1470 non-null   int64
 25  RelationshipSatisfaction  1470 non-null   int64
 26  StandardHours             1470 non-null   int64
 27  StockOptionLevel          1470 non-null   int64
 28  TotalWorkingYears         1470 non-null   int64
 29  TrainingTimesLastYear     1470 non-null   int64
 30  WorkLifeBalance           1470 non-null   int64
 31  YearsAtCompany            1470 non-null   int64
 32  YearsInCurrentRole        1470 non-null   int64
 33  YearsSinceLastPromotion   1470 non-null   int64
 34  YearsWithCurrManager      1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
```

[8]:
```
data.describe()
```

[8]:
```
                Age    DailyRate  DistanceFromHome    Education  EmployeeCount  \
count  1470.000000  1470.000000       1470.000000  1470.000000         1470.0
mean     36.923810   802.485714          9.192517     2.912925            1.0
std       9.135373   403.509100          8.106864     1.024165            0.0
min      18.000000   102.000000          1.000000     1.000000            1.0
25%      30.000000   465.000000          2.000000     2.000000            1.0
50%      36.000000   802.000000          7.000000     3.000000            1.0
75%      43.000000  1157.000000         14.000000     4.000000            1.0
max      60.000000  1499.000000         29.000000     5.000000            1.0

       EmployeeNumber  EnvironmentSatisfaction   HourlyRate  JobInvolvement  \
```

count	1470.000000	1470.000000	1470.000000	1470.000000
mean	1024.865306	2.721769	65.891156	2.729932
std	602.024335	1.093082	20.329428	0.711561
min	1.000000	1.000000	30.000000	1.000000
25%	491.250000	2.000000	48.000000	2.000000
50%	1020.500000	3.000000	66.000000	3.000000
75%	1555.750000	4.000000	83.750000	3.000000
max	2068.000000	4.000000	100.000000	4.000000

	JobLevel	JobSatisfaction	MonthlyIncome	MonthlyRate \
count	1470.000000	1470.000000	1470.000000	1470.000000
mean	2.063946	2.728571	6502.931293	14313.103401
std	1.106940	1.102846	4707.956783	7117.786044
min	1.000000	1.000000	1009.000000	2094.000000
25%	1.000000	2.000000	2911.000000	8047.000000
50%	2.000000	3.000000	4919.000000	14235.500000
75%	3.000000	4.000000	8379.000000	20461.500000
max	5.000000	4.000000	19999.000000	26999.000000

	NumCompaniesWorked	PercentSalaryHike	PerformanceRating \
count	1470.000000	1470.000000	1470.000000
mean	2.693197	15.209524	3.153741
std	2.498009	3.659938	0.360824
min	0.000000	11.000000	3.000000
25%	1.000000	12.000000	3.000000
50%	2.000000	14.000000	3.000000
75%	4.000000	18.000000	3.000000
max	9.000000	25.000000	4.000000

	RelationshipSatisfaction	StandardHours	StockOptionLevel \
count	1470.000000	1470.0	1470.000000
mean	2.712245	80.0	0.793878
std	1.081209	0.0	0.852077
min	1.000000	80.0	0.000000
25%	2.000000	80.0	0.000000
50%	3.000000	80.0	1.000000
75%	4.000000	80.0	1.000000
max	4.000000	80.0	3.000000

	TotalWorkingYears	TrainingTimesLastYear	WorkLifeBalance \
count	1470.000000	1470.000000	1470.000000
mean	11.279592	2.799320	2.761224
std	7.780782	1.289271	0.706476
min	0.000000	0.000000	1.000000
25%	6.000000	2.000000	2.000000
50%	10.000000	3.000000	3.000000
75%	15.000000	3.000000	3.000000

```
max            40.000000               6.000000         4.000000

       YearsAtCompany  YearsInCurrentRole  YearsSinceLastPromotion  \
count     1470.000000         1470.000000              1470.000000
mean         7.008163            4.229252                 2.187755
std          6.126525            3.623137                 3.222430
min          0.000000            0.000000                 0.000000
25%          3.000000            2.000000                 0.000000
50%          5.000000            3.000000                 1.000000
75%          9.000000            7.000000                 3.000000
max         40.000000           18.000000                15.000000

       YearsWithCurrManager
count           1470.000000
mean               4.123129
std                3.568136
min                0.000000
25%                2.000000
50%                3.000000
75%                7.000000
max               17.000000
```

Observations

we only have int and string data types features. there is no feature with float. 26 features are numerical and 9 features are categorical Attrition in out target value which has no missing value. But, the quantity of data of emp having Attrition is less compared to employees whoch do not have Attrition. It's very good that we are having a complete dataset, there is no any missing values in data set.

[9]:
```
print(data.duplicated().value_counts())
data.drop_duplicates(inplace = True)
print(len(data))
```

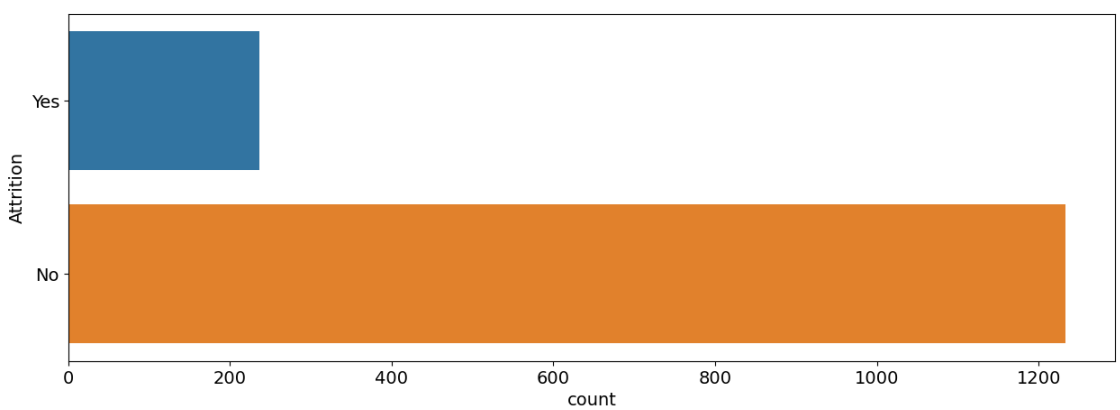
```
False    1470
Name: count, dtype: int64
1470
```

[10]:
```
data.isnull().sum()
```

[10]:
```
Age                         0
Attrition                   0
BusinessTravel              0
DailyRate                   0
Department                  0
DistanceFromHome            0
Education                   0
EducationField              0
```

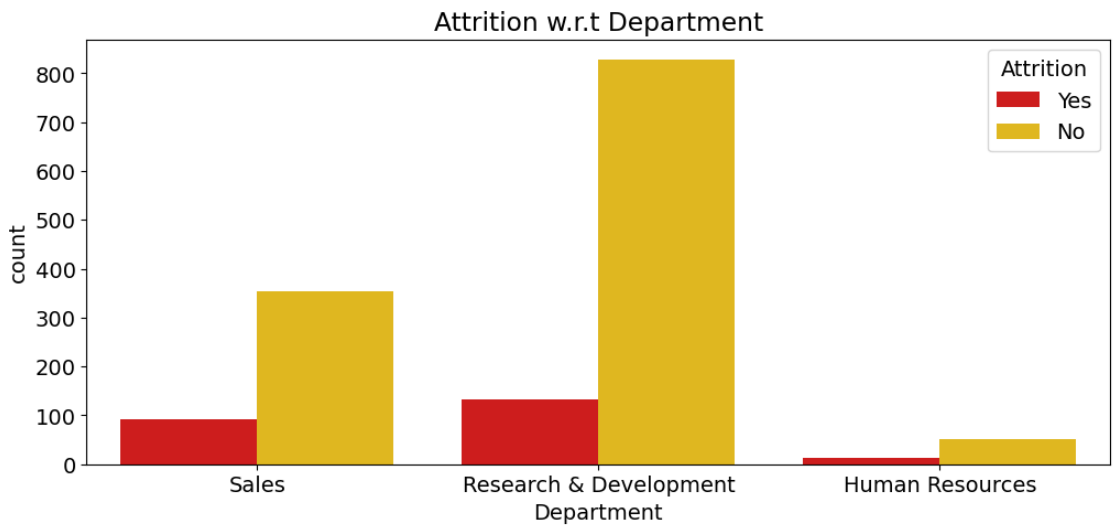
```
EmployeeCount               0
EmployeeNumber              0
EnvironmentSatisfaction     0
Gender                      0
HourlyRate                  0
JobInvolvement              0
JobLevel                    0
JobRole                     0
JobSatisfaction             0
MaritalStatus               0
MonthlyIncome               0
MonthlyRate                 0
NumCompaniesWorked          0
Over18                      0
OverTime                    0
PercentSalaryHike           0
PerformanceRating           0
RelationshipSatisfaction    0
StandardHours               0
StockOptionLevel            0
TotalWorkingYears           0
TrainingTimesLastYear       0
WorkLifeBalance             0
YearsAtCompany              0
YearsInCurrentRole          0
YearsSinceLastPromotion     0
YearsWithCurrManager        0
dtype: int64
```

[11]:
```
plt.figure(figsize=(15,5))
plt.rc("font", size=14)
sns.countplot(y ='Attrition',data=data)
plt.show()
```

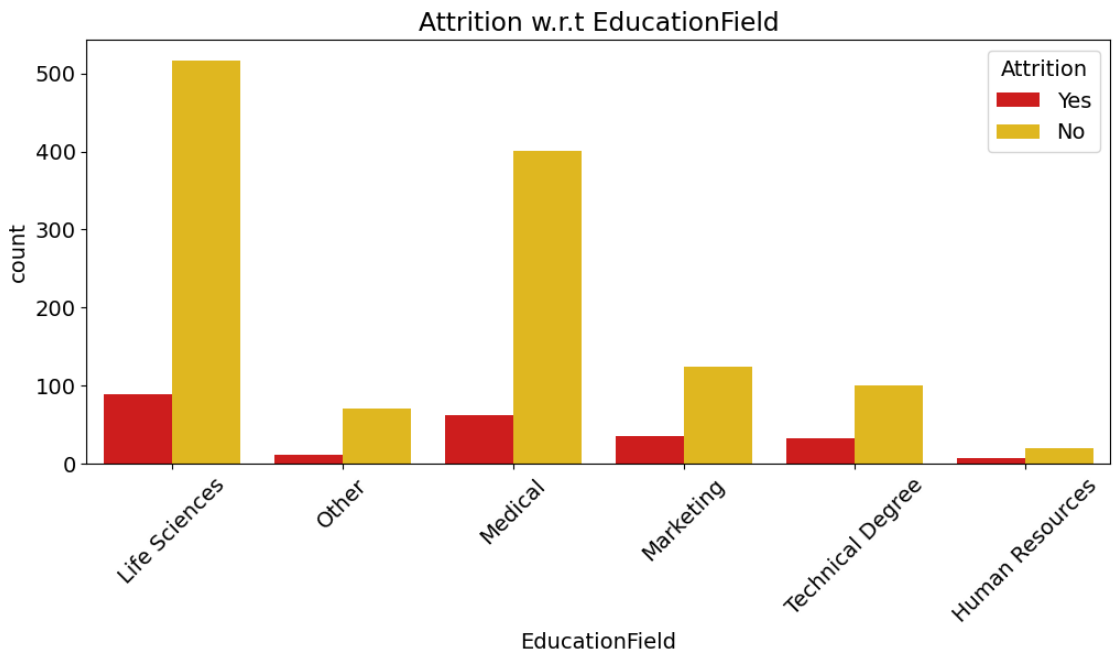


Over here we noticed that the Target column is Highly Imbalanced, we need to balance the data by using some Statistical Methods. Exploratory Data Analysis

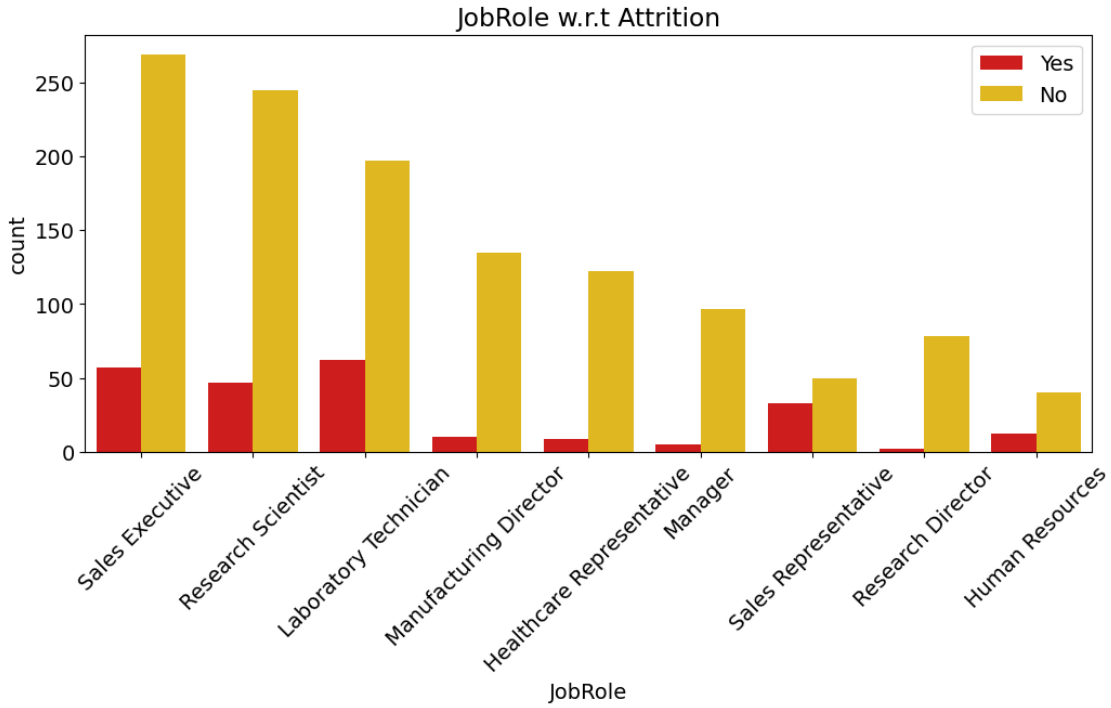
[12]:
```
# Department wrt Attrition
plt.figure(figsize=(12,5))
sns.countplot(x='Department',hue='Attrition', data=data, palette='hot')
plt.title("Attrition w.r.t Department")
plt.show()
```



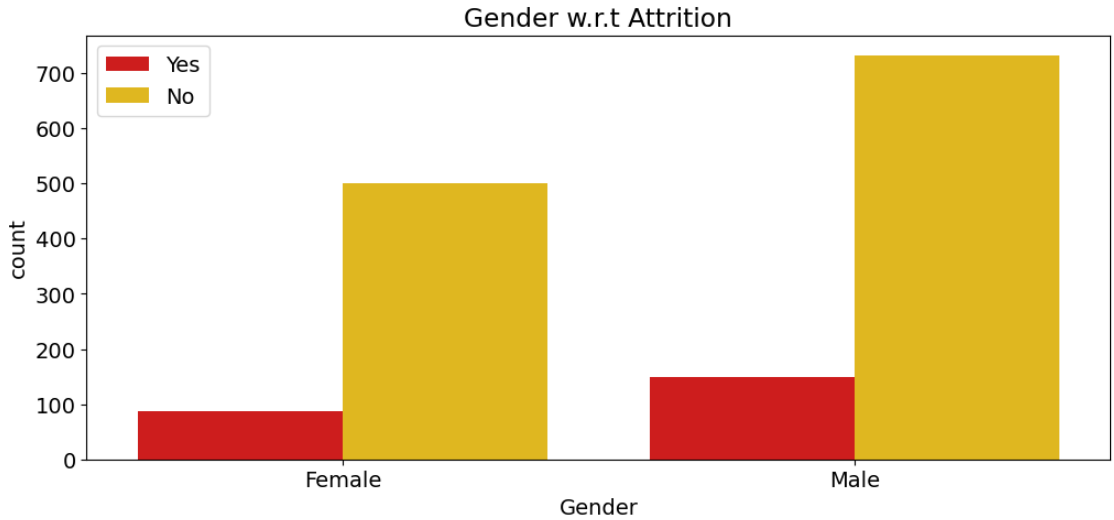
[13]:
```
# Department wrt Attrition
plt.figure(figsize=(12,5))
sns.countplot(x='EducationField',hue='Attrition', data=data, palette='hot')
plt.title("Attrition w.r.t EducationField")
plt.xticks(rotation=45)
plt.show()
```



[14]:
```
# let's see at which post most people are leaving the jobs
# JobRole
plt.figure(figsize=(12,5))
sns.countplot(x='JobRole',hue='Attrition', data=data, palette='hot')
plt.title("JobRole w.r.t Attrition")
plt.legend(loc='best')
plt.xticks(rotation=45)
plt.show()
```



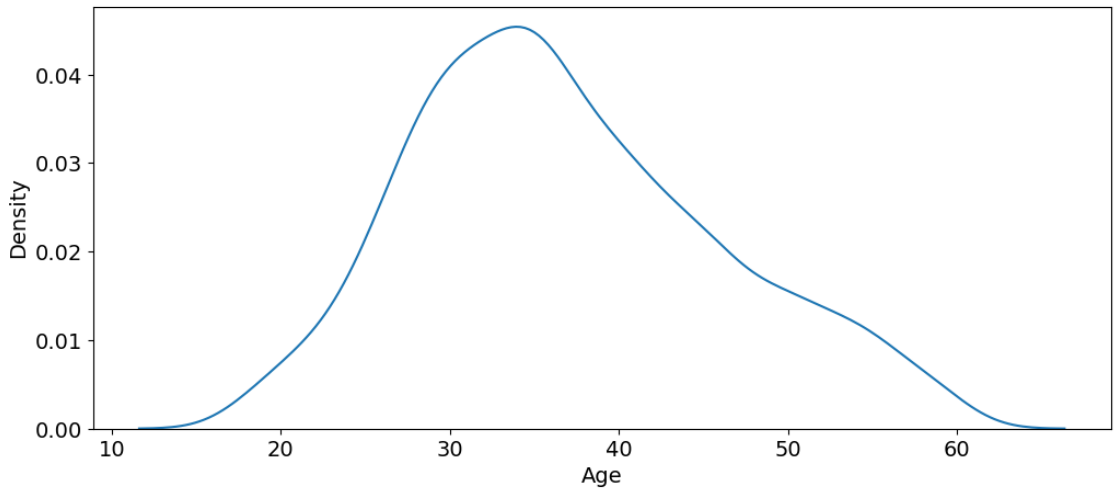
[15]:
```
# most male of female employes Attriate
# Department wrt Attrition
plt.figure(figsize=(12,5))
sns.countplot(x='Gender',hue='Attrition', data=data, palette='hot')
plt.title("Gender w.r.t Attrition")
plt.legend(loc='best')
plt.show()
```



OBSERVATIONS

Employees working in R&D department are more, but employees from sales department or at position like sales executive,sale Representative leaves the job early. Males are more under Attrition then Females

[16]:
```
# distribution of age
plt.figure(figsize=(12,5))
sns.distplot(data['Age'],hist=False)
plt.show()
```

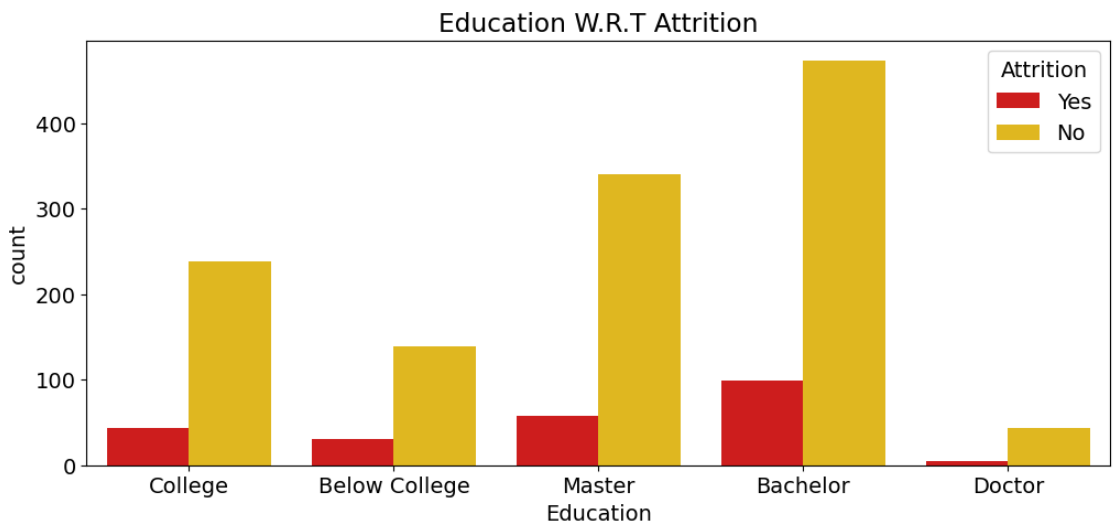


[17]:
```
ordinal_features =␣
↪['Education','EnvironmentSatisfaction','JobInvolvement','JobSatisfaction',
                    ␣
↪'PerformanceRating','RelationshipSatisfaction','WorkLifeBalance']
data[ordinal_features].head()
```

[17]:
```
   Education  EnvironmentSatisfaction  JobInvolvement  JobSatisfaction  \
0          2                        2               3                4
1          1                        3               2                2
2          2                        4               2                3
3          4                        4               3                3
4          1                        1               3                2

   PerformanceRating  RelationshipSatisfaction  WorkLifeBalance
0                  3                         1                1
1                  4                         4                3
2                  3                         2                3
3                  3                         3                3
```

```
4                    3                              4                  3
```

[18]:
```
edu_map = {1 :'Below College', 2: 'College', 3 :'Bachelor', 4 :'Master', 5:␣
↪'Doctor'}
plt.figure(figsize=(12,5))
sns.countplot(x=data['Education'].map(edu_map), hue='Attrition', data=data,␣
↪palette='hot')
plt.title("Education W.R.T Attrition")
plt.show()
```



OBSERVATIONS

Employees from Bachelor are more, then from Masters background. Attrition wrt to bachelor can be seem more because they have more and more expectation from companies and it will be interesting to see the reason behind this in this dataserds.

[19]:
```
# Target Variable(Attrition)
data['Attrition'] = data['Attrition'].replace({'No':0,'Yes':1})
```

[20]:
```
#encode binary variables
data['OverTime'] = data['OverTime'].map({'No':0,'Yes':1})
data['Gender'] = data['Gender'].map({'Male':0,'Female':1})
data['Over18'] = data['Over18'].map({'N':0,'Y':1})
```

[21]:
```
data.head()
```

[21]:
```
   Age  Attrition     BusinessTravel  DailyRate              Department  \
0   41          1      Travel_Rarely       1102                   Sales
1   49          0  Travel_Frequently        279  Research & Development
2   37          1      Travel_Rarely       1373  Research & Development
```

```
3   33          0  Travel_Frequently       1392  Research & Development
4   27          0      Travel_Rarely        591  Research & Development

   DistanceFromHome  Education EducationField  EmployeeCount  EmployeeNumber  \
0                 1          2  Life Sciences              1               1
1                 8          1  Life Sciences              1               2
2                 2          2          Other              1               4
3                 3          4  Life Sciences              1               5
4                 2          1        Medical              1               7

   EnvironmentSatisfaction  Gender  HourlyRate  JobInvolvement  JobLevel  \
0                        2       1          94               3         2
1                        3       0          61               2         2
2                        4       0          92               2         1
3                        4       1          56               3         1
4                        1       0          40               3         1

                JobRole  JobSatisfaction MaritalStatus  MonthlyIncome  \
0        Sales Executive                4        Single           5993
1     Research Scientist                2       Married           5130
2  Laboratory Technician                3        Single           2090
3     Research Scientist                3       Married           2909
4  Laboratory Technician                2       Married           3468

   MonthlyRate  NumCompaniesWorked  Over18  OverTime  PercentSalaryHike  \
0        19479                   8       1         1                 11
1        24907                   1       1         0                 23
2         2396                   6       1         1                 15
3        23159                   1       1         1                 11
4        16632                   9       1         0                 12

   PerformanceRating  RelationshipSatisfaction  StandardHours  \
0                  3                         1             80
1                  4                         4             80
2                  3                         2             80
3                  3                         3             80
4                  3                         4             80

   StockOptionLevel  TotalWorkingYears  TrainingTimesLastYear  \
0                 0                  8                      0
1                 1                 10                      3
2                 0                  7                      3
3                 0                  8                      3
4                 1                  6                      3

   WorkLifeBalance  YearsAtCompany  YearsInCurrentRole  \
0                1               6                   4
```

```
1                 3              10                   7
2                 3               0                   0
3                 3               8                   7
4                 3               2                   2

   YearsSinceLastPromotion  YearsWithCurrManager
0                        0                     5
1                        1                     7
2                        0                     0
3                        3                     0
4                        2                     2
```

[22]:
```
df = pd.DataFrame(data)
```

[23]:
```
one_hot_encoded = pd.get_dummies(df)
```

[24]:
```
print(one_hot_encoded)
```

```
      Age  Attrition  DailyRate  DistanceFromHome  Education  EmployeeCount  \
0      41          1       1102                 1          2              1
1      49          0        279                 8          1              1
2      37          1       1373                 2          2              1
3      33          0       1392                 3          4              1
4      27          0        591                 2          1              1
…   …         …         …                 …         …              …
1465   36          0        884                23          2              1
1466   39          0        613                 6          1              1
1467   27          0        155                 4          3              1
1468   49          0       1023                 2          3              1
1469   34          0        628                 8          3              1

      EmployeeNumber  EnvironmentSatisfaction  Gender  HourlyRate  \
0                  1                        2       1          94
1                  2                        3       0          61
2                  4                        4       0          92
3                  5                        4       1          56
4                  7                        1       0          40
…              …                        …     …          …
1465            2061                        3       0          41
1466            2062                        4       0          42
1467            2064                        2       0          87
1468            2065                        4       0          63
1469            2068                        2       0          82

      JobInvolvement  JobLevel  JobSatisfaction  MonthlyIncome  MonthlyRate  \
0                  3         2                4           5993        19479
1                  2         2                2           5130        24907
```

2	2	1	3	2090	2396
3	3	1	3	2909	23159
4	3	1	2	3468	16632
…	…	…	…	…	…
1465	4	2	4	2571	12290
1466	2	3	1	9991	21457
1467	4	2	2	6142	5174
1468	2	2	2	5390	13243
1469	4	2	3	4404	10228

	NumCompaniesWorked	Over18	OverTime	PercentSalaryHike	\
0	8	1	1	11	
1	1	1	0	23	
2	6	1	1	15	
3	1	1	1	11	
4	9	1	0	12	
…	…	…	…	…	
1465	4	1	0	17	
1466	4	1	0	15	
1467	1	1	1	20	
1468	2	1	0	14	
1469	2	1	0	12	

	PerformanceRating	RelationshipSatisfaction	StandardHours	\
0	3	1	80	
1	4	4	80	
2	3	2	80	
3	3	3	80	
4	3	4	80	
…	…	…	…	
1465	3	3	80	
1466	3	1	80	
1467	4	2	80	
1468	3	4	80	
1469	3	1	80	

	StockOptionLevel	TotalWorkingYears	TrainingTimesLastYear	\
0	0	8	0	
1	1	10	3	
2	0	7	3	
3	0	8	3	
4	1	6	3	
…	…	…	…	
1465	1	17	3	
1466	1	9	5	
1467	1	6	0	
1468	0	17	3	
1469	0	6	3	

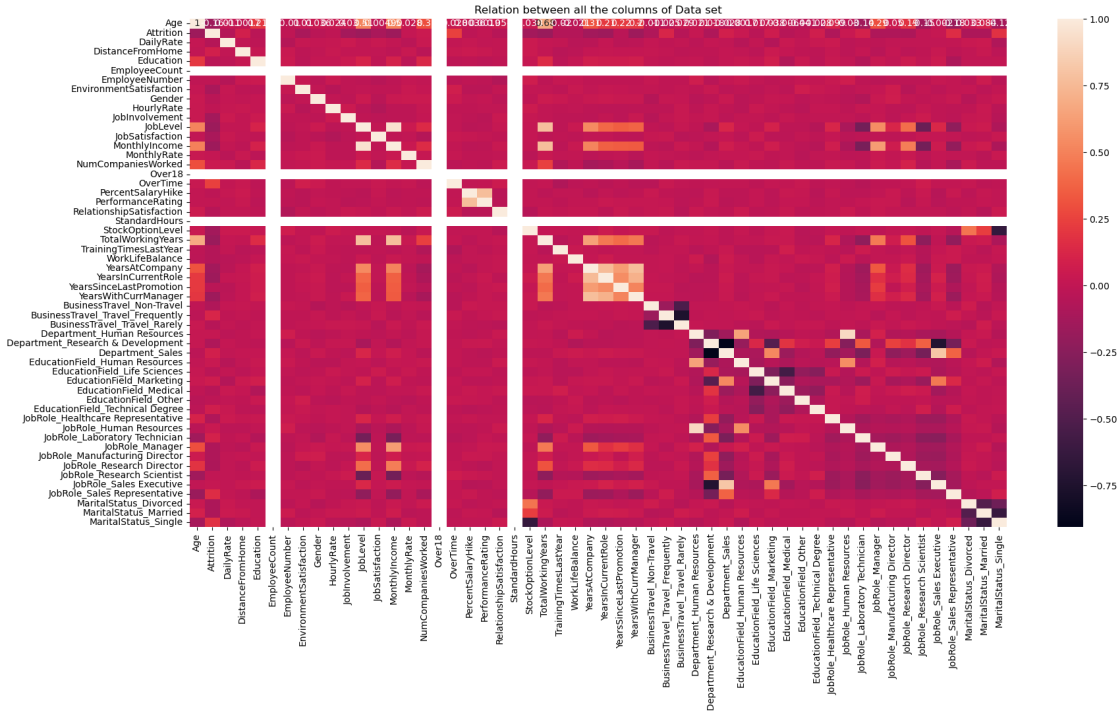
```
      WorkLife Balance  YearsAtCompany  YearsInCurrentRole  \
0                    1               6                   4
1                    3              10                   7
2                    3               0                   0
3                    3               8                   7
4                    3               2                   2
…                  …               …                   …
1465                 3               5                   2
1466                 3               7                   7
1467                 3               6                   2
1468                 2               9                   6
1469                 4               4                   3

      YearsSinceLastPromotion  YearsWithCurrManager  \
0                           0                     5
1                           1                     7
2                           0                     0
3                           3                     0
4                           2                     2
…                         …                     …
1465                        0                     3
1466                        1                     7
1467                        0                     3
1468                        0                     8
1469                        1                     2

      BusinessTravel_Non-Travel  BusinessTravel_Travel_Frequently  \
0                         False                             False
1                         False                              True
2                         False                             False
3                         False                              True
4                         False                             False
…                         …                                 …
1465                      False                              True
1466                      False                             False
1467                      False                             False
1468                      False                              True
1469                      False                             False

      BusinessTravel_Travel_Rarely  Department_Human Resources  \
0                             True                       False
1                            False                       False
2                             True                       False
3                            False                       False
4                             True                       False
…                            …                         …
1465                         False                       False
```

```
1466                          True                       False
1467                          True                       False
1468                         False                       False
1469                          True                       False

      Department_Research & Development  Department_Sales  \
0                                 False              True
1                                  True             False
2                                  True             False
3                                  True             False
4                                  True             False
…                                  …                …
1465                               True             False
1466                               True             False
1467                               True             False
1468                              False              True
1469                               True             False

      EducationField_Human Resources  EducationField_Life Sciences  \
0                              False                          True
1                              False                          True
2                              False                         False
3                              False                          True
4                              False                         False
…                              …                             …
1465                           False                         False
1466                           False                         False
1467                           False                          True
1468                           False                         False
1469                           False                         False

      EducationField_Marketing  EducationField_Medical  EducationField_Other  \
0                        False                   False                 False
1                        False                   False                 False
2                        False                   False                  True
3                        False                   False                 False
4                        False                    True                 False
…                        …                       …                     …
1465                     False                    True                 False
1466                     False                    True                 False
1467                     False                   False                 False
1468                     False                    True                 False
1469                     False                    True                 False

      EducationField_Technical Degree  JobRole_Healthcare Representative  \
0                               False                              False
1                               False                              False
2                               False                              False
```

```
3                              False                              False
4                              False                              False
…                              …                                …
1465                           False                              False
1466                           False                               True
1467                           False                              False
1468                           False                              False
1469                           False                              False

      JobRole_Human Resources  JobRole_Laboratory Technician  JobRole_Manager  \
0                       False                          False            False
1                       False                          False            False
2                       False                           True            False
3                       False                          False            False
4                       False                           True            False
…                       …                              …               …
1465                    False                           True            False
1466                    False                          False            False
1467                    False                          False            False
1468                    False                          False            False
1469                    False                           True            False

      JobRole_Manufacturing Director  JobRole_Research Director  \
0                              False                      False
1                              False                      False
2                              False                      False
3                              False                      False
4                              False                      False
…                              …                        …
1465                           False                      False
1466                           False                      False
1467                            True                      False
1468                           False                      False
1469                           False                      False

      JobRole_Research Scientist  JobRole_Sales Executive  \
0                          False                     True
1                           True                    False
2                          False                    False
3                           True                    False
4                          False                    False
…                          …                      …
1465                       False                    False
1466                       False                    False
1467                       False                    False
1468                       False                     True
1469                       False                    False
```

```
      JobRole_Sales Representative  MaritalStatus_Divorced  \
0                            False                   False
1                            False                   False
2                            False                   False
3                            False                   False
4                            False                   False
…                            …                     …
1465                         False                   False
1466                         False                   False
1467                         False                   False
1468                         False                   False
1469                         False                   False

      MaritalStatus_Married  MaritalStatus_Single
0                     False                  True
1                      True                 False
2                     False                  True
3                      True                 False
4                      True                 False
…                      …                    …
1465                   True                 False
1466                   True                 False
1467                   True                 False
1468                   True                 False
1469                   True                 False

[1470 rows x 54 columns]
```

[25]:
```
plt.figure(figsize=(20,10))
plt.rc("font", size=10)
sns.heatmap(one_hot_encoded.corr(),annot=True)
plt.title("Relation between all the columns of Data set")
plt.show()
```



Machine Learning Algorithm

[26]:
```
one_hot_encoded.isnull().sum()
```

[26]:

Age	0
Attrition	0
DailyRate	0
DistanceFromHome	0
Education	0
EmployeeCount	0
EmployeeNumber	0
EnvironmentSatisfaction	0
Gender	0
HourlyRate	0
JobInvolvement	0
JobLevel	0
JobSatisfaction	0
MonthlyIncome	0
MonthlyRate	0
NumCompaniesWorked	0
Over18	0
OverTime	0
PercentSalaryHike	0
PerformanceRating	0

```
RelationshipSatisfaction             0
StandardHours                        0
StockOptionLevel                     0
TotalWorkingYears                    0
TrainingTimesLastYear                0
WorkLifeBalance                      0
YearsAtCompany                       0
YearsInCurrentRole                   0
YearsSinceLastPromotion              0
YearsWithCurrManager                 0
BusinessTravel_Non-Travel            0
BusinessTravel_Travel_Frequently     0
BusinessTravel_Travel_Rarely         0
Department_Human Resources           0
Department_Research & Development    0
Department_Sales                     0
EducationField_Human Resources       0
EducationField_Life Sciences         0
EducationField_Marketing             0
EducationField_Medical               0
EducationField_Other                 0
EducationField_Technical Degree      0
JobRole_Healthcare Representative    0
JobRole_Human Resources              0
JobRole_Laboratory Technician        0
JobRole_Manager                      0
JobRole_Manufacturing Director       0
JobRole_Research Director            0
JobRole_Research Scientist           0
JobRole_Sales Executive              0
JobRole_Sales Representative         0
MaritalStatus_Divorced               0
MaritalStatus_Married                0
MaritalStatus_Single                 0
dtype: int64
```

[27]:
```
X = one_hot_encoded.drop(['Attrition'], axis=1)
y = one_hot_encoded['Attrition'].values
```

[28]:
```
# Split the data into training and testing set
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,␣
↪random_state=42)
```

[29]:
```
# Sanity check for the sampled data
print(X_train.shape)
print(y_train.shape)
```

```
print(X_test.shape)
print(y_test.shape)
```

```
(1176, 53)
(1176,)
(294, 53)
(294,)
```

[30]:
```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, accuracy_score, roc_curve,␣
↪roc_auc_score
```

[31]:
```
logreg = LogisticRegression()
logreg.fit(X_train, y_train)
```

[31]:
```
LogisticRegression()
```

[32]:
```
prediction=logreg.predict(X_test)
cnf_matrix = confusion_matrix(y_test,prediction)
print("Accuracy Score is", accuracy_score(y_test , prediction)*100)
```

```
Accuracy Score is 86.73469387755102
```

[ ]:
```

```